GDP 2015 Final Report

IRG-07 Front Wing Design

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Abstract

Designing an aerodynamic package to accommodate the change to smaller diameter tyres was the objective set at the start of the project. The front wing optimised for this objective produced 166N of downforce at 15ms^{-1} and was a major contributor to the total of 366N of downforce produced by the whole aerodynamic package. High strength carbon fibre composite with a layup of $[0°/0°/\pm45°/90°]_S$ and a thickness of 2mm was used in a sandwich structure along with low density foam to constitute the front wing. The front wing weighed a total of 7.6kg and the material cost equals £370. A CFD tool was used to optimise individual parameters of the front wing and also to evaluate design choices. Given a larger time span, wind tunnel data should be used to further validate and correlate the CFD code. An optimisation strategy that accounted for two varying parameters would also produce a better estimate of the optimum, although a trade off exists as this would be more computationally expensive.

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Introduction

The 2015 Imperial Racing Green (IRG) Group Design Project (GDP) aims at producing a viable design, fully compliant with the FSAE rules, which the IRG team can potentially use to enter the Formula Student competition in the future. Therefore, one of the main drivers has been the attention to costs, trying to reuse as many components as we can from the team while attempting to achieve the best result in the race.

This included reusing IRG's current electric package and building up on the latest chassis they have designed, an aluminium and carbon fibre space frame, which has evolved into a light yet reliable semi-monocoque concept. The GDP team has also ensured that most if not all the components can be acquired from a manufacturer and need not be custom made. This places significant constraints and design limitations but the team has managed to produce an affordable design for EV04 with an improved performance from its predecessors thanks to a reduction in weight as well as a redesigned aerodynamic package and the use of smaller wheels while keeping some features such as an electric all-wheel drive and the usage of torque vectoring.

Global vehicle layout

A key aspect of car design is the initial sizing process. In order to maximise the vehicle's performance, a series of preliminary weight, centre of gravity and stability calculations were effected with the relevant teams. The baseline configuration was constrained by the width of the existing front wing used by the Imperial Racing Green team, which has been developed further during this project, and FSAE regulations. The wheelbase was minimised and track width was maximised accordingly to increase stability, and were defined as 1525 and 1255 millimetres respectively.

The next step was the positioning of all the various components and assessing the required connections between all of them. The vehicle layout was arranged around the pilot. The cockpit was designed to satisfy the minimal cross-sectional area and shape FSAE requirements. Special attention is paid to the location of the centre of gravity (CG), which is important in determining the handling characteristics of a racing vehicle. For EV4, 3 key considerations were taken into account during the design phase: firstly, for a 4 Wheel Drive (4WD), equal loading in each wheel would maximise traction and this can be achieved with a 50:50 front-to-rear weight distribution. In addition to this, by placing heavier objects as close to this point as possible, a low moment of inertia (MoI) will ensure quicker steering response. Finally, a CG position close to the ground will reduce dynamic load transfers, which is especially important since we have chosen the minimum allowable wheelbase length.

In order to achieve said configuration and since the pilot has a fixed position in the chassis, the two significantly heavy components other than the pilot (the battery pack and the electric package containing all four motor controllers) were split between the front and the back. Additional components, such as the water pumps and water tank were also added at the front to palliate the weight difference between the battery pack, which was placed at the rear, and the electric package. In order to keep the centre of gravity as low as possible and to allow attachment to a floor, the pilot was inclined as much as the cockpit dimensions allowed and the components placed as closely to the bottom of the chassis as the constraints permitted. Based on the initial design layout, a CG position was determined by approximating each component into simple geometric shapes and assumed to be uniform in density. This value was found to be at 42.4% of the wheelbase and 245mm from the ground. Although the longitudinal position deviated by 116mm from the initial target, it provided an estimate for stability calculations. The battery pack was subsequently repackaged and moved rearwards in an attempt to achieve the target weight distribution.

The previous EV03 design used 20-inch tyres with 13-inch rims, but we have decided to downsize to 16-inch tyres with 10-inch rims in order to explore the benefits of smaller wheels as proposed on the design brief. The choice of 10-inch wheel rims came down to the limitations of packaging the brake system, motors and transmission into the wheels. The choice of 16-inch tyres came down to the smallest readily available tyres which we could find to fit the 10-inch rims, which are also the closest approximation in size to the specimens used by testers and the rest of the teams providing sufficient data for the wheels to be analysed as smaller tyres do not fit the rigs used by testers and therefore no data is available. The most suitable tyres found are the very soft compound 16-inch radial tyres from Avon motorsport.

The wheel packaging was the most consequent challenge, as the wheel's size was to be minimised while it had to contain the motor, gearbox, brakes, cooling system, sensors and a casing to shield the assembly. The most difficult task was to integrate the cooling to the casing, which was realised with the addition of fins, and the connections of the wires from the chassis to the wheel. However, this greatly reduced the number of components that had to be placed in the chassis. Wheels and suspensions were successfully integrated onto the chassis.

Last stages of the packaging process included the design of the cooling system and its layout around the whole car, as well as the electric wiring. Sensors and pilot controls (excluding the steering wheel and pedal box) were the latest additions. The diffuser clearance was also found to be 14.6mm more than the minimum ground clearance. This resulted in a successful weight distribution, with a CG placed 48.5% from front of wheelbase and an increased height of 288mm, which could be lowered with an improved battery and motor controller design, along with a redesign of the seat.

Weight

The nature of the course and dynamic events, specifically the endurance and acceleration tests, favours vehicles with good acceleration capabilities, and therefore the desire for a lightweight vehicle design. Furthermore, the Formula student competition has no minimum weight requirement, so the weight reduction benefits essentially have no bound. A decision making process was undertaken for component selection considering performance, weight and cost for all the components on the car. The weights of most components were estimated to a reasonable accuracy using the CAD models, whilst the remaining components were estimated by contacting the manufacturers.

The mass estimate of the car is 217.4kg excluding the driver. We have made a conservative weight estimate for the endurance event drivers of about 68kg, which gives a combined mass of car including driver of 285.4kg. This results in EV04 being lighter (around 15kg) than its predecessor, partly the reason of its improved performance in the dynamic events.

Cost and Sustainability

In a project so constrained by the availability of investors, costs become extremely important. For the development of the EV04 project, an initial price estimate showed an approximate cost of $35000, excluding manufacturing, software and other external costs. After more in depth analysis and contact with providers, a more exact cost of $43297 was found. The main components of this cost are the batteries, the motors and the bodywork. In the case of the batteries these are set and the only possibility of price change would be a reduction in the number of cells. It must be noted that this design incorporates the current battery pack the Racing Green team already has and a carbon fibre and aluminium space frame which they will be reusing in order to reduce significant costs. The motors were chosen to be high end to ensure high performance and an improvement in results from previous years. The bodywork has the most space for improvement in both cost and weight, with a reduction in size being recommended by the current team.

Furthermore, a business case was conducted, in which an analysis of additional costs, were we to produce the vehicle as a manufacturer, was carried out. This investigation yielded interesting results, concluding that due to the niche market and the high initial costs, the best way to perform would be as a subsidiary of one or various large vehicle manufacture firms (such as aircraft or road cars) as a tester of different new technologies.

The sustainability of the vehicle is extremely important when considering that the Formula Student competition encourages the use of electric vehicles. Although an exact carbon emission figure is impossible to achieve at this stage of the design, an initial approximation showed that the car would emit approximately 200g CO_2/km, including both manufacture and running of the vehicle, a figure much smaller than would be for an internal combustion engine car of similar characteristics.

Compliance with regulations

There is a long list of tests to be passed as well as many obscure loading situations which need to be analysed and put into a report before the vehicle can race. Most of these loading cases and tests will not be straightforward and will require knowledge that spans many degree disciplines. Some document submissions have a very low pass rate for the 1st/2nd try and as a result time and effort can be lost and major changes needed to be implemented if the vehicle is not adherent in some way. During the project design choices were influenced by the required time of regulation submissions, and this was the key reason for choosing to use the standard Impact Attenuator given by FSAE. Designing an Impact Attenuator in-house would require two pages of lengthy testing procedures be carried out. Another reason why an understanding of the regulation is important is because, this competition is not simply about engineers building a car that is fast and manoeuvres well. There is a fine intermingling between such elements as the costing of the vehicle, its efficiency that will go make up about 40% of the points available in the competition which are related to non-engineering aspects of vehicle design.

Regulation most heavily touches the insulation layers needed by the car. Of which there are several different types (i.e.) firewall. The chassis structure is also heavily regulated and has to satisfy requirements for frontal impact and side impact. A good understanding of the regulation will allow the selection of design choices which can garner a higher number of points, and reduce the time and cost of the vehicle (which in this competition is also equivalent to points). There is a large interplay between elements of the car and many elements will influence each other, and a good understanding of the regulations is required because often times a regulation on one part of the car will affect a completely unrelated structure of the car.

Aerodynamics

The aerodynamic package is responsible for producing downforce to help provide better traction. This is especially helpful when cornering, as a greater speed can be taken into the corner, resulting in more favourable times. As shown by the lap simulator, the aerodynamic package provides a noticeable advantage in the endurance event.

Front wing

The front wing is a major contributor to the whole package and is also responsible for directing the required mass flow to the side pods for efficient cooling and also to the diffuser. The design of the front wing is heavily influenced by the new set of regulations introduced this year. A height restriction placed in front of the tyres calls for the higher elements of the front wing to be truncated. Instead a single flap is placed here with an angle optimised to direct flow over the tyres most efficiently. The main plane is continuous for the whole span of the front wing and has a 5 degree angle of attack. Three elements have also been used wherever possible in-order to maximise downforce.

The front wing has a total span of 1435 mm and extends a total of 700 mm from the leading edge of the front tyre. No part of the front wing exists in the zone of 80mm just in front of the tyres to allow the tyre to be completely unobstructed. Eppler 421 and Eppler 420 were used for the main plane and the flaps, due to the high maximum lift coefficient and large stall angles.

Once the outline of the basic design was established, CFD and empirical data were used in order to optimise the ground clearance, angles of attack, slot gaps and endplates. Following optimisation studies that varied a single parameter, it was established that the two most influential parameters were the main plane angle of attack and the ground clearance. Downforce was most sensitive to any variations in these parameters. Therefore to find a better optimum the decision to carry out two parameter studies was made.

Using results produced by optimisation studies and physical arguments the design of the from wing was decided upon. A ground clearance of 65 mm with a main plane angle of 5 degrees produced the highest downforce. The main flap angles were 39 and 45 degrees whilst the flap directly in front of the tire was placed at 40 degrees. Slot gaps of 10 mm with a 5 mm overlap were also chosen.

Structural considerations were also made throughout the design process. Using a sandwich construction the front wing was tested against the maximum possible bending moment. Following the use of conservative estimates, a layup of $[0°/0^{circ}/\pm 45^{circ}/90^{circ}]_S$ produced deflections much smaller than the allowable limit. Using this layup produced a total thickness of 2mm resulting in a total weight of 7.6 kg. Seeing as the produced deflections are well below the allowable limit, reduction of the laminate is possible in order to reduce both weight and cost.

Rear wing

Initially, a downforce of 200N was specified to maintain balance with the front wing and diffuser. This had to be generated while keeping inside the geometric constraints set by the Formula Student rules: max span is set at 1100mm, max height at 1200mm and it cannot be placed forward of the drivers rear support nor more than 250mm behind the rearmost face of the rear tire.

There were several compromises that had to be made in terms of the design based on the downforce and drag generated by the wing and the overall weight of the design. Drag was taken as being an insignificant factor, as the motors would be capable of overcoming the increase in drag that usually came with any increase in downforce. However, the weight was considered to be a significant factor in the performance of the wing, this resulted in the aim of the design being to generate the required 200N at the minimum possible weight.

The initial concept for the wing was a single wing-plane with an inverted three element design, a main aerofoil with a leading and trailing edge high-lift device. The aerofoils E71 for the main element and S1223 for the leading edge and trailing edge devices. Both are highly chambered which gives them high downforce at the low speed at which the car functions and have very high CLmax, the values of which were found using x-foil data. Side plates were added to increase the effective aspect ratio of the wing and manage the flow over the wing. Flap gaps and overlaps were set at 2% main wing chord and the flap itself was set as 60% of main wing chord. Initial simulations of this design in free stream conditions revealed that it generated only 90N of downforce for 20N of drag, nowhere near the target for balance. This resulted in some major changes to the design. Due to the size of the leading edge device it would be difficult to produce and had very little effect on the downforce, it was therefore abandoned.

A second wing plane was also added as theoretically it would nearly double the downforce and drag, but the increase in drag was deemed to be worth the doubled downforce. After determination and validation of a mesh, CFD simulations were run in Star CCM+ with the bodywork, driver and chassis included. This gave a large increase in the number of required cells, however it was necessary to include these effects as the wing was fully immersed in the wake of these objects, so they would have a significant effect on the downforce. These simulations gave an overall downforce of 170N and drag of 54.4N, which is a much better result in terms of balance, however it did involve nearly doubling the weight to 9.68 kg. Due to time constraints this design was deemed to be sufficient, as further optimisation of the design only gave a shift of around 5-10N. The double wing plane also performs better than a single wing plane (1.5 points more) based on the lap simulator. This means that the double wing performs only slightly better than the single wing at a large increase in weight and cost.. However, the double wing plane is selected since the higher downforce is better for the dynamic balance of the car.

Two concepts were considered for the support structure of the rear wing. The first was four struts to the rear portion of the chassis secured with aluminium fasteners. The second was to extend the side plates down to the chassis. Both of these designs required new bars to the rear of the chassis, which resulted in a fairly significant increase in the weight of the structure. In the end the first concept was chosen, as the side plate supports were found to be unfeasible as the thickness of plate required is very large and even then it would provide now lateral stability. This resulted in the basic strut design having a lower weight that outweighed the negative effects that central struts had on the downforce.

Several further improvements that could still be made to the wing include experimenting with different wing devices, such as gurney flaps or a second trailing edge device. It would also be worth investigating removing any wing section immersed in the wake of the car, as this has the potential to give a large weight saving without having too large a negative effect on the down force generated.

Bodywork

The objective of the bodywork is to first cover the internals of the car, fit over the chassis and produce an aerodynamic efficient body that does not reduce the capabilities of the front wing and hence the car overall as well as minimising the drag produced. A number of iterations were carried to help optimise the design with priorities given to reducing weight, attachment to the front wing and making sure the wheelbase and hence the suspension was not affected. The final mass was given as 11.2kg.

Therefore, the bodywork was initially designed to first meet the regulations and fit a 95th percentile male and 5th percentile female. A high nose concept was also chosen after conversations with global vehicle layout, front wing and diffuser team to help give more space for the internals such as the electric package and maximise downforce for the front wing and to allow flow to go downwards towards the diffuser and hence maximise the downforce further. Bodywork of length 2200mm and 666mm height is produced. The maximum width with the side pods is found to be 1084mm.

The objective of the sidepods is to maximise the cooling for the radiators and is designed by the given parameters of the cooling team. The sidepods are also coordinated with the undertray which cannot have a smaller width than the sidepods. Simulations are used with the radiator in the sidepod after interaction with the front wing as the radiators act as a blockage and hence will cause drag. With the radiator dimensions and inlet area given, the design of the sidepods was completed with a constant area for 354mm till after the radiator where the sidepod converges to help reduce mass and concentrate the flow.

The objective of the rear cover is to protect the batteries from debris or in case of wet conditions. Therefore, the main priority is to minimise the mass as there are no structural requirements of the rear cover as all attachments such as suspension and the rear wing are joined to the chassis which will take the load. However, drag can be reduced by making the rear cover more aerodynamic which will be a larger design but with a lighter material it will offset that disadvantage. Therefore, the rear cover is designed to not only minimise mass but also reduce drag via the separation that can be caused at the rear.

The final values from the full car simulation show that the bodywork is producing a lift of 13.8N and drag of 46.5N however the priority of the bodywork is to minimise mass and help facilitate the front wing and the diffuser.

Diffuser

The underbody is considered a low drag method of generating downforce on a race car. The function of an undertray is to increase the velocity of the air underneath the car, therefore reducing the static pressure and thus producing downforce in order to give the tires more grip and allowing higher acceleration. The diffuser is designed to expand and decelerate the underbody airflow thus reducing the back pressure on the flow under the car in order to satisfy the free stream conditions.

Several Formula Student rules regarding the dimensions and position of the undertray and diffuser affect the design significantly. In addition, a jacking bar of minimum length of 300mm must be visible from the rear of the car, so this was mounted on the diffuser. In order to minimise weight, the under tray was shaped to the bodywork and side pods and underwent several length reductions until the final design weighing 4.93 kg was reached.

CFD simulations were conducted using the STAR-CCM+ software package in order to analyse the aerodynamics effects of undertray and diffuser design. The results of the simulations were in a form of pressure distribution along the body, the results suggesting 40mm of ground clearance and a diffuser angle of 12 degree would produce the most downforce. By conducting a full car simulation, the final design suggested it would produce 44.2N of downforce and 3.0N of drag at 15m/s, which is the average velocity of the vehicle in the endurance dynamic event.

Cooling

The electric vehicle we have come up with involves the use of a variety of components that would be found in a modern formula one car, which operate with inefficiencies in the form of heat. The process of cooling helps alleviate the system from overheating by drawing heat away and allowing it to dissipate into the surroundings such that it can remain in operation, at least for the duration of the race.

The set of rules imposed by the FSAE regulations needed to be met. The system was modelled to use plain water as from specifications of the battery it shouldn't operate above a temperature of 55°C, in this case significant stresses will not arise from temperature constraints in the pipes housing the fluid. The cooling system also needs a means to store the fluid and for this we will be using catch tanks. The catch tanks were selected to contain 0.9L of water. They will also be situated in such a position below the driver's shoulder level as outlined on the regulations. They are made from aluminium, capable of containing boiling water without deformation, and can vent water out of the car once the race is over, towards the bottom of the frame.

Meeting these requirements, the maximum heat loss in the battery and motor controllers were found, and a liquid cooling system has been designed around this. As the motors were directly exposed to an airflow slightly outside the wheel, and operated at ≈90% efficiency they didn't require a liquid cooling system and could be instead cooled sufficiently from the oncoming airflow.

The liquid cooling system is comprised of two radiators housed in the side pods, and aluminium plates for heat transfer between the battery pack. The motor controllers have their own heat sink and only require a water supply for their cooling. Radiator dimensions were then determined, as well as liquid mass flow rates and temperature changes across the battery, and were used to calculate a required air intake area and mass flow rate into the side pods. The water will be circulated using water pumps, and will be supplied with water from the catch tank. Temperature sensors can be used to monitor temperature rises across the motor controllers and battery and used to regulate water flow in the housing.

A model was created to estimate the heat build up in the brake discs around the circuit including the effect of cornering, a cooling model was similarly proposed and used to estimate a surface area required of an air cooling duct that could be provided to the brakes to cool them to an allowable limit. Parts were then found from readily available motorsport markets, and were chosen in similar dimensions to those required by calculation.

Structures

Starting from the fundamental approach of hybrid space-frame and composites panel assembled bodywork, the chassis was designed as the mounting body that carries all main payloads including the driver, batteries, and the electric packages. The body panels were divided in to 15 pieces in total, 13 as [+45/-45]2s laminates to maintain the aerodynamic shape of the body, and 2 half-nosecone shells of [0, 90, +45/-45]2s panels due to the further structural requirement on the nose section as it was designed to carry the 7kg front wing generating a maximum down force of 438 N, thus bending moment along the nosecone.

The space-frame was simulated in Finite Element Analysis regarding to the maximum dynamic loading case during racing where the car experiences bumping when cornering with full braking applied. Results have shown that the maximum chassis displacement stays within the constrained deformation in the aspect of ground clearance and to maximise the effect on aerodynamic performance. Regarding to the results of the failure prediction using the Maximum Failure Criterion, the design was justified with non-failure performance in both static and dynamic events.

For the final design, the chassis was constructed using UD carbon fibre tubes with aluminium inserts at jointing points, covered by IM7/8552 carbon fibre/Epoxy composites panels. The estimated cost of the chassis is 2800 GBP and that of the panels, 620 GBP.

Suspension and vehicle stability

The FSAE rules and regulations state that the car must be equipped with a fully operational suspension system with shock absorbers in the front and rear. After ensuring rules and regulations are set to be passed, the focus on suspension design shifts to both the suspension geometry, and shock absorber and dampener design. This is to provide both good ride and handling performance.

The suspension geometry was designed with a short-long arms (SLA) system, incorporating pull rods into both the rear and front which attach to the springs and dampeners in the form of rockers. The tie rod themselves attach to the chassis by a joint with circular ends which can be tightened in the form of bolts to ensure rigidity. It is designed such that all the wheels can house the motors and transmission for all conceivable wheel travels and subsequent change in geometry. The upright provides mounting for the suspension wishbones, the brakes, the transmission, the tie rods and also the steering rod in the front case. It is designed to give minimal displacement under loading whilst also minimising the mass. Due to the limited time the design is kept simple so future work could be done to minimise the mass further.

A case of roughly 60% anti-dive and anti-squat are achieved for both the front and rear suspension. This allows for less wheel travel during any longitudinal load transfer as most of the loading is taken by the wishbones. This has been managed by arranging the geometry such that we achieve a front roll centre of 25% the height of the centre of gravity and the rear as 15% of the height of the centre of gravity. These roll centres are found to be a reasonable compromise of trying to minimise both the rolling moment and the jacking moment.

In order to attain a ground clearance of 40mm – beneficial to the diffuser design – a front ride frequency of 6 Hz was chosen, with the rear ride frequency 10% less than this. This resulted in desired spring rates of 139 N/mm for the front springs and 90 N/mm for the rear springs.

A wheel loading model was created that incorporated the weight distribution between front and rear wheels, the downforce from the aero package, and the subsequent load transfer during braking at 1.92g, accelerating at 0.9g and cornering at 1.75g. A dynamic loading factor of 2 was used to account for the effects of the vehicle hitting a bump. The maximum front wheel loading was 3355 N on the outside wheel, whilst cornering and braking at 19m/s. The maximum rear wheel loading was 2611 N on the outside wheel, whilst cornering and accelerating at 19m/s.

Taking into account the effects of anti, the maximum wheel travel under loading was 39.0 mm. Subsequently, an Avo Universal Twin Tube Coilover damper was chosen, with double adjustability and length of 279.4 mm open and 203.2 mm closed, passing the regulation stipulating a minimum usable wheel travel of ±25.4 mm. 152.4 mm Faulkner racing springs were then chosen, of 140.3 N/mm and 92.1 N/mm respectively. The damper and spring constitute a shock absorber, these were placed as a result of the geometry of the vehicle and its components.

A linear Elementary Automobile model with 2 degrees of freedom was created to model the vehicle during steady state cornering. Due to the lack of data from Avon for the chosen tyres, the model uses cornering stiffnesses interpolated from larger tyres. Stability derivatives and linear steady state control response characteristics were calculated, and a number of methods, including a positive stability factor, showed the car to be very slightly understeer. The neutral steer point was found to be a non-dimensionalised 48.8% of the wheelbase from the front tyres, resulting in a static margin of 0.38%, again suggesting the vehicle is close to neutral steer, as with most race cars. The tangent speed of the car was calculated to be 9.5 m/s and the characteristic speed was 108 m/s - unattainable for the vehicle in any case. The steering sensitivity was found to increase with velocity and using data from the lap simulator, a graph of time in lap vs steering wheel angle was created, that would give the driver an idea of how best to handle the car through the corners, assuming the transient stages were negligible. On a track day, through changes in tyre pressure and subsequently cornering stiffness, the vehicle could be set up as understeer or oversteer, depending on the ability of the driver.
A model of transient stability and control analysed the sideslip response to a lateral force and the yaw response to steer angle. As velocity increased, the response time to reach 90% steady state increased for both cases. The yaw response time to 90% steady state at the maximum cornering velocity, 19 m/s, is 0.26 seconds. The damping factor at this speed is 0.99, essentially it is critically damped and the vehicle will return to a steady state with little overshoot.

Electrical Systems

The goal of the electrical system and devices is the relay information from the sensors to the processors, and from the processors to the controllers, which will regulate the amount of power required by the motor during different drive conditions such as turning, braking, or accelerating by regulating the voltage and current input.

The controller is the main component used to regulate the input current and voltage to the motors. When choosing the controller, the main concern is the maximum operating current and voltage the motor runs on. The motor is chosen by the Motor Selection Team is the JM1 and runs at a maximum of 350V and current 50A. The controller is then chosen to be at 360V with current 60A. Another concern when it comes to controller is the weight of the controller. For this amount of voltage and current, industrial controllers are easily found in the market. However, industrial controller are significantly heavy, an average of 8kg. This is why all top ranked teams have self-manufactured controllers, some weighing only 1kg depending on the setup. The final choice made was a controller manufactured by EV Craft, which weighs only 2.147 kg each. Seeing that all wheel drive is the set up for our current design, 4 controllers are needed for all 4 motors, therefore, a weight save of 18.612kg by using the self-manufactured controller compared to the average industrial controller. The controller has a CAN bus installed in it to allow communication between the controllers.

An Arduino Due microcontroller board acts as an intermediate between all the sensors and motors.

The sensors record the car conditions and translate them as signals to the Arduino Due. The Arduino can be then programmed to read these signals and translate them into torque value and fed to the controller via the CAN Bus. The controller will then translate that information into amount of current required and distribute the current correctly.

The driving system of the car is a combination of the torque vectoring system and the potentiometer sensor which functions as the torque encoder. When the gas pedal is travelled, the potentiometer senses the amount of travel and sends the information via the CAN bus to the Arduino microprocessor. Since the motor is capable of using either its peak torque value or continuous torque value, the microprocessor will process the optimum amount of torque required without slipping and thus sends the signal of the optimum combination of current and voltage to the microcontrollers. The microcontrollers then sends the current and voltage to the motors which then runs the motors, which then runs the car.

Braking System

The goal of the brake system is to brake the vehicle from the maximum performance speed with the largest brake force and gives the best braking "feel" of the car. From the lap simulator, the larger the brake force, the better the performance. Therefore, an iterative method is used to choose the components of the braking system which gives the maximum amount of brake force of 1.92g. Most components are bought from AP Racing, a reliable source for buying brake components and are used by a lot of other FSAE competitors and have shown reliable results. This iterative method is conducted such that whenever the rear wheel locks before the front wheel locks, the iteration stops. This is because the front wheel should lock first before the rear wheel for safety reasons. Front wheel locking is more stable and safer when it comes to human over brake error. Rear wheel locking is dangerous as it makes the car unstable and hard to control.

When choosing the brake components, another important aspect other than maximising the amount of brake force is the weight of the components. Since this is a 4 wheels drive configuration, most of the weight of the brake components will be in the wheels. The magnitude of sprung mass of the car affects the performance and stability of the car greatly. The lower the sprung mass, the better the stability and performance. Therefore, low weight material such as stainless steel and motorbike brakes which weight less than car brakes are used. Lastly, the size of the rim is also limiting the brake disc size. Compared to previous years where the rims are larger than this year's design, the disc brake which are to fit in the rim are limited as well. Another design is would be to build the brake disc outside of the wheels, therefore increasing the brake disc size to a larger amount. This is also very beneficial in terms of cooling as well. However, due to restriction in time when this design was discovered, we did not go on with the idea.

Steering System

The goal of the steering system is to steer the car through the track. The steering angle is adjusted depending on the steepest angle when cornering, which is 51.3°. By using a safety factor of 20%, the maximum steering angle of the tire is 61.6°. The steering ratio is the ratio of how far you turn the steering wheel to how far the wheels turn. This steering ratio is currently set as 0.77 and can be adjusted based on the preference of the driver. The current steering ratio setup is so that the driver can perform the maximum steering of the car without crossing his or her hands.

There are two types of steering mechanism, mechanical steering and hydraulic steering. Mechanical steering is generally lighter, cheaper and simpler than hydraulic steering. Therefore mechanical steering is used. Within mechanical steering, there are two types, worm drive and rack and pinion steering. Rack and pinion is simpler to design and lighter than worm drive steering, however more expansive. The final decision was made to use rack and pinion as the weight save and simplicity was significant enough to comprehend the cost.

Power-train

During the five dynamic events of the Formula student competition, the powertrain should ideally be capable of delivering the maximum torque that could possibly be transferred to the ground by the tires. An over-engineered design however, with more motor torque than can be transferred for a certain track and tire condition, would result in the wheels slipping and also implies a weight penalty. On the other hand, an under-capacity in terms of torque would yield a lower race performance.

Motors and batteries

A study of motor requirements was conducted by considering torque and power estimates, wheel rim dimensional constraints, FSAE rules and dynamic events as well as battery pack constraints. A motor fulfilling all the requirements was not sourced on the market as an off-the-shelf product. Only five Formula Student teams were found to have an in-wheel system with 4 motors, which were all self-developed or sponsored. A constraint of liquid cooling was removed as it is a non-trivial solution that requires a very detailed and complex analysis.

The most suitable electric motor on the market was found to be Jobymotors' 2.7 5 kg Brushless DC JM1. Characteristic performance parameters are a continuous and peak power of 13.2 kW and 20.1 kW respectively as well as a continuous and peak torque of 21 Nm and 32 Nm. The four in-wheel motors are connected in parallel to the battery pack, which consists of 84, 0.428 kg EiG C020 Lithium Ion cells providing a capacity of 6.1 kWh.

A custom winding option of 360 V was selected to comply with the given 50-60 kg battery pack with a maximum charge voltage of 348.6 V. The maximum current of battery pack is 200 A, yielding a maximum power of 69.72 kW. The battery pack can therefore not power the four motors at peak power simultaneously. However, the JM1 motor was still preferred over the less powerful and lighter motor version JM1S. When running the Lap Simulator, the JM1 yields a 20 second faster race time in the most important event in terms in terms of points, the endurance event.

Temperature heavily affects the motor performance and therefore the maximum possible surface area of the motor casing was designed, to alleviate air cooling, while still fitting between the suspension arms. The 0.69 kg aluminium casing was also designed to be waterproof, serve as scatter protection and resist transmission loads when the motor runs at peak torque.

Drive-train

The purpose of drive-train mechanical design is to ensure delivery of power from motor to wheel. Components involved in the design are gearboxes, adaptors, shafts, bearings, shaft seals, bolts, nuts and washers. The challenge of in-wheel drive design is that all the parts must be compact and be able to fit into the space in the wheel rim.

The gear ratio is decided based on optimum motor and car performance. Forces acting on wheels were used for structural analysis on all the components.

In the end the APEX DYNAMICS 1 stage 7:1 ratio AE090 planetary gearbox is chosen. The output adaptor, input shaft and output shaft are designed to fit the specific in-wheel geometrical configuration. Bearings and shaft seals are chosen from SKF and all the bolts, nuts and washers can be purchased from RS.

In the end, the drive-train components on each wheel have a total mass of 4.03kg and cost about $285.84 (excluding cost of human labour, transport, bolts, nuts and washers).

Traction Control and Torque Vectoring

Traction Control and Torque Vectoring systems work in tandem to improve the application of thrust through the tyres and increase vehicle stability when cornering. Traction Control eliminates excessive wheel spin which maximises the amount of torque transferred to the road surface. A Torque Vectoring system dynamically alters the individual torques at each wheel; done by changing motor voltages in our 4 motor system. This enables an assisting yaw moment to be generated as the vehicle manoeuvres, allowing it to remain stable. Both systems use the same components to create a closed feedback loop to remove uncertainty. Raw data is collected by various sensors on the vehicle and a microcontroller calculates the error between the actual values and desired values. Using a CAN bus, the required correction factors are transmitted to the motor controllers and consequently the motor voltages are changed. In order to validate the use of these systems, the lap simulator was used; the lap times with and without the systems were estimated and the penalty in lap times due to the added weight of the components was calculated. Overall, the decrease total lap time in the endurance event was found to be 59.67 seconds or 2.71 seconds faster per lap which is significant enough to justify the use of this system. The estimated increase in score due to torque vectoring is approximately 48 points; considering the very low cost and weight of the system and the benefits it brings to the car, more research should be done to optimise this as much as possible. Additionally, work could be done to improve the reliability of the systems as electrical systems are more susceptible to failure than mechanical ones.

Lap simulator and vehicle performance

The Lap Simulator is an important tool to quickly and reliably assess the outcome of changing design parameters. It accepts inputs from a large number of aspects of the vehicle, including geometrical dimensions, aerodynamic and mechanical forces. The current Lap Simulator that has been used for assessing the design was originally developed by the 2014 IRG Group, but significant modifications has been made throughout the 2015 IRG Project. The vehicle model has been upgraded from a two-dimensional bicycle model to a three-dimensional car model, and a braking system has been introduced to generate more convincing results. The modified Lap Simulator now provides good estimations of finish times for all four dynamic events of the Formula Student competition: Acceleration, Sprint (Autocross), Skid Pad, and Endurance. It also compares the estimated times with the results from Formula Student 2014, to predict the possible scores and places the Imperial Racing Green Team could achieve with the design of EV04 from the 2015 IRG Project.

The predicted performance of the final design of EV04 are shown as below:

Table 1: Predicted Performance of EV04 from Lap Simulator

Event	Time (s)	Score (points)	Place
Acceleration	4.0	40.87	6
Skid Pad	4.6	50.00	1
Sprint (Autocross)	56.2	76.91	15
Endurance	1455.0	285.80	2

Due to the way the existing lap simulator was written it proved very difficult to improve the tyre model. Therefore a different and more function based approach was used to write a new version. With this method each system and subsystem in the race was modelled using a function that would then be easily upgradeable given enough time. The advantage of this is that it allows a far greater number of variables to be accounted for in the simulation, so that aspects such as steering geometry and torque vectoring can be analysed. Also the accuracy and complexity of the simulation can easily be increased by adding new system functions or improving the models within the existing functions.

The top level models are the vehicle model, the track model and the driver model. The vehicle model also contains all the vehicle subsystem model function such as the tyre model. The overall dynamics model was derived assuming a 6-DOF rigid body vehicle, however including the assumption of no rotation about longitudinal and lateral axes and no vertical velocity this reduces to 3-DOF. The track model takes the racing line x-y coordinates and interprets them into distance-curvature coordinates to be passed to the driver model. The driver model produces the accelerator, braking, gear and steering inputs to be used in the vehicle model based on the outputs of the track and vehicle models.

To complete this new lap simulator the vehicle model must be verified by comparing its output in certain situations to expected values obtained from analytical calculations such as for straight acceleration or constant speed cornering. The driver model also remains to be implemented and most model functions could be further improved.

Conclusions

All in all, EV04 has been developed to a high standard considering the time constraints of this project. Its performance has been improved from its predecessors, placing consistently in the top ten teams in the competition. Weight has been reduced from previous designs, as well as the overall cost, which has been considered a key driver throughout the whole process. A more accurate lap simulator and vehicle stability model have also contributed to the success of this design. The vehicle incorporates existing components such as the space frame and battery pack and new ideas like the carbon fibre panels to form the bodywork in a hybrid concept and a sophisticated aerodynamic package. There is however room for improvement, as the battery pack should be redesigned for efficiency and mass saving purposes and the rear aerodynamic package could be simplified for the same reason.

Front Wing Report

Introduction

Racing green is Imperial College London's contribution to the Formula Student competition that has contestants competing from around the world. This year as part of our group design project we were tasked with producing a conceptual design for an open wheel race car to compete in 2016. 'EV4', the iteration of design that we are working on, will be an all electric vehicle that will compete under class 1 of the Formula Student rules.

This report aims to outline the design procedure undertaken whilst highlighting key decisions taken and problems encountered during the course of the project. The directives applying to the design of the front wing and therefore the ones that heavily influenced our design approach are reported in the section below.

Project directive

- Incorporate the use of smaller wheels
- Design a new aerodynamics package to accommodate the new wheels
- Comply with new regulations

The aerodynamic package that includes the front wing is responsible for producing downforce. This is essential as it allows for higher cornering speeds due to added traction and therefore leading to a faster car. In order to investigate the necessity of an aerodynamic package, the lap simulator was used to asses the time advantage that could be gained. The figure below depicts this relationship.

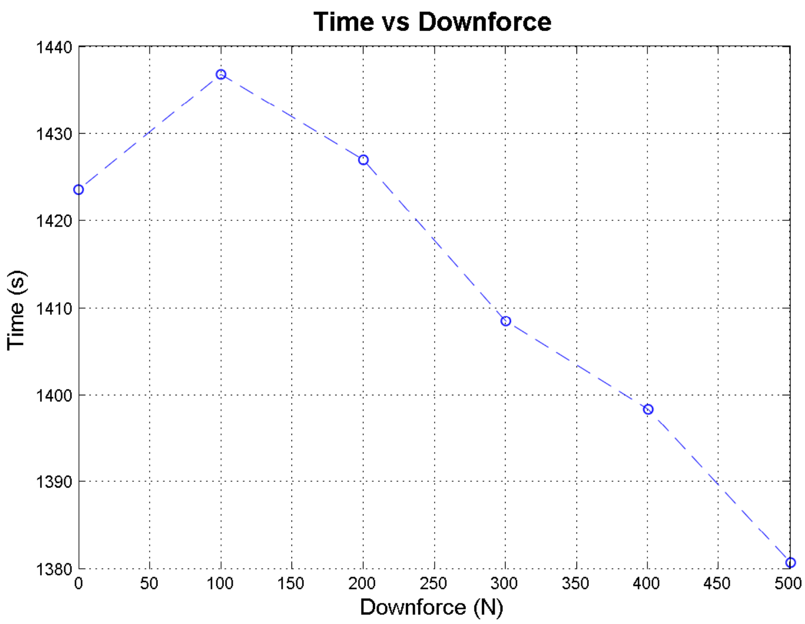


Figure 1: Variation in total time with downforce

It can be seen that a marked improvement is made when an aerodynamic package is installed. Initially the added downforce is not favourable as the added weight outweighs the advantage of producing downforce. However, when the car produces more than 200N of downforce an aerodynamic package is justified. From this point onwards higher the value of the downforce the faster the car will be over the course of the event.

The front wing is an essential part of the whole aerodynamic package and is responsible for producing a large proportion of downforce produced by the car. However the front wing should also be designed to ensure sufficient mass flow is directed into the under-tray of the car which is channelled to the diffuser. Another design constraint is to facilitate efficient cooling. This calls for high momentum air to be directed towards the inlet of the side-pods, where the radiator is located.

The main design driver is not just limited to maximising downforce, but also to reduce weight of the whole structure. To understand the importance of the latter, the lap simulator was used to evaluate performance when, either the aerodynamic forces or the mass was kept constant whilst the other varied by 5% (mass was reduced while aerodynamic forces were increased). This allowed us to conclude that a mass reduction reduced overall times by 1.2% whereas increasing the aerodynamic forces (both lift and drag) provided a much smaller improvement in total time. These times are valid for the endurance event. The endurance event is used as a reference as it is most valuable in terms of allocated points.

Regulations

Compliance to regulations is of critical importance to all parts of this project as a design that doesn't comply with the FSAE regulations will result in disqualification.[1] Major changes in regulations with reference to the front wing hugely influenced the new design. Figure 2 describes the area in which any aerodynamic devices can be placed.

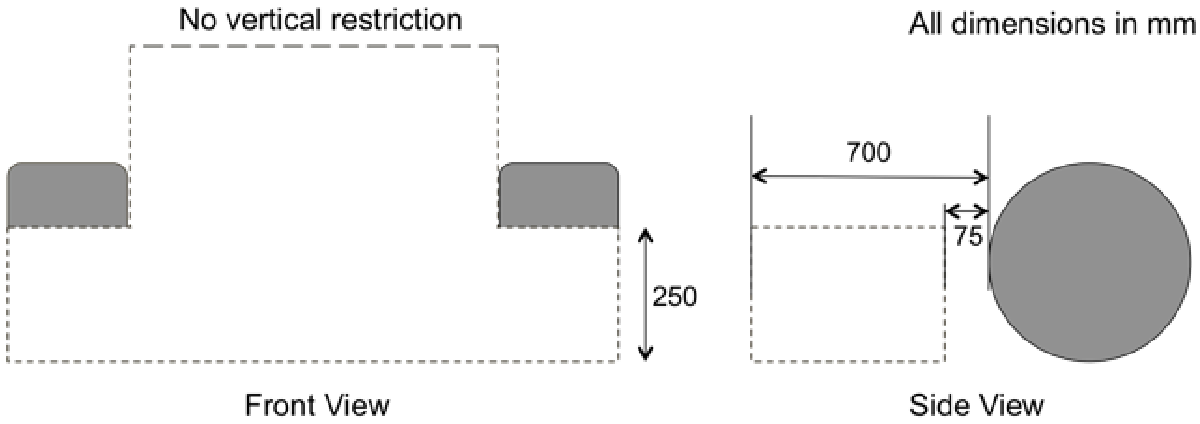


Figure 2: FSAE regulations regarding the front wing (Not to scale)

No vertical restriction is set for the placement of any aerodynamic devices in between the two tyres. The restricted 75mm zone directly in front of the tyre ensures that the tyre is unobstructed. A notable exception to figure 2 is that any vertical surfaces such as endplates with a thickness of less than 25mm do not need to comply with the height restriction set directly in front of the tyres.

Design

The change in tyre size from 20 inches to a 16 inch outer diameter heavily influenced the initial design process. Apart from being an objective set for the project a change to smaller tyres reduced the bluff body plan form area and therefore reduced drag. Initial simulations showed the drag to decrease by 13% due to the change in tyre size alone. These simulations were done on an unoptimised wing design that was inherited from Imperial Racing Green. Such a choice of starting point was justified as it provided us with initial data upon which to build on.

The 250mm restriction set in front of the tyres meant that the shape of the wing cannot be continuous along the span. As can be seen in figure 3 the higher elements of the front wing are truncated directly in front of the tyre. Three elements have also been used wherever possible in order to maximise downforce and therefore improve times.(see figure 1) Eppler 421 and Eppler 420 aerofoils have been used for the main plane and flaps respectively. The choice of aerofoil was inspired by previous work done by the other part of the front wing sub-team in collaboration with Imperial Racing Green. These aerofoils had a high $C_{L_{max}}$ and large stall angles, therefore they were ideal for use in the front wing.

The endplates chosen have been optimised for weight. Structural capabilities have also been considered. Please note that the detailed procedure to optimise each parameter and the strategy used has been covered later in the report. Figure 3 provides the CAD model for the final design.

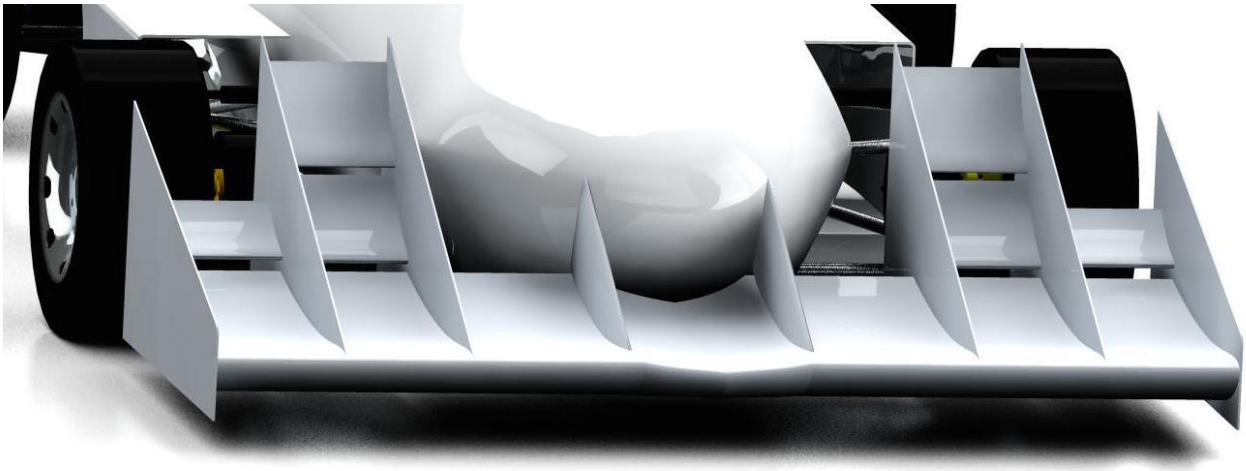


Figure 3: CAD model of the final design

Computational fluid dynamics

CFD was a tool that was used most during the course of this project in order to evaluate our design decisions. Starccm+ is the CFD tool we used to optimise each part of our design. Table 2 highlights the physical models used to simulate flow around the front wing. Notably, segregated flow and the K-ϵ turbulence model have been used.

Table 2: Physics models used in CFD simulations

Physics models	
Steady	Three dimensional
Gas	Constant density
All y+ wall treatment	Segregated flow
Turbulence	K-ϵ turbulence model

Both the segregated flow and the coupled flow model were compared before the segregated flow model was chosen to be used. The most relevant reasons include, when using the coupled flow model the number of iterations needed for convergence to achieved was found to be larger than when using the segregated flow model. This is also backed up by theory as the number of iterations required by the coupled flow solver is independent of the mesh size whereas convergence in the segregated flow case depends on the size of the mesh. Limitations to the segregated flow model arise when the solver is required to account for compressible effects or when required to solve problems in the supersonic regime. However, as this doesn't concern us, we decided to choose the segregated flow solver.

Polyhedral meshes were used over hexahedral and tetrahedral meshes due to the faster convergence rate and robust convergence at lower residual values.[2] Mesh size for simulations including only the front wing and surrounding components resulted in approximately 1.2 million cells. Whereas the full car simulation produced 6.1 million cells. Figure 5 displays an example of the mesh used and it can clearly be seen that volumetric controls have been used to refine the mesh in sensitive areas such as near the wing and in the wake region. Volumetric control allows a more refined mesh to be produced allowing for a more accurate calculation but the total cell size isn't increased greatly as a much coarser mesh surrounding the fine region exits.

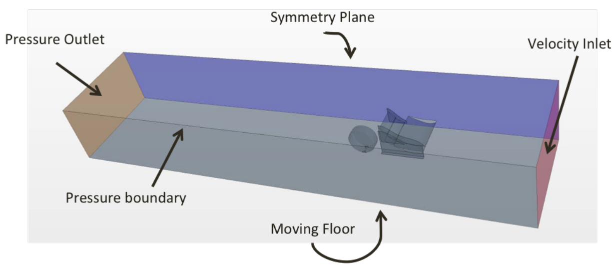


Figure 4: Boundary conditions used in CDF simulations

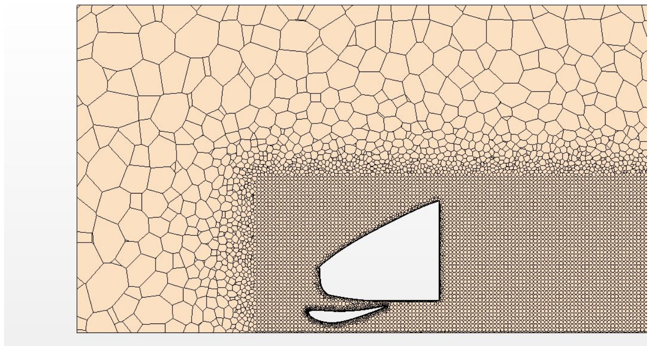


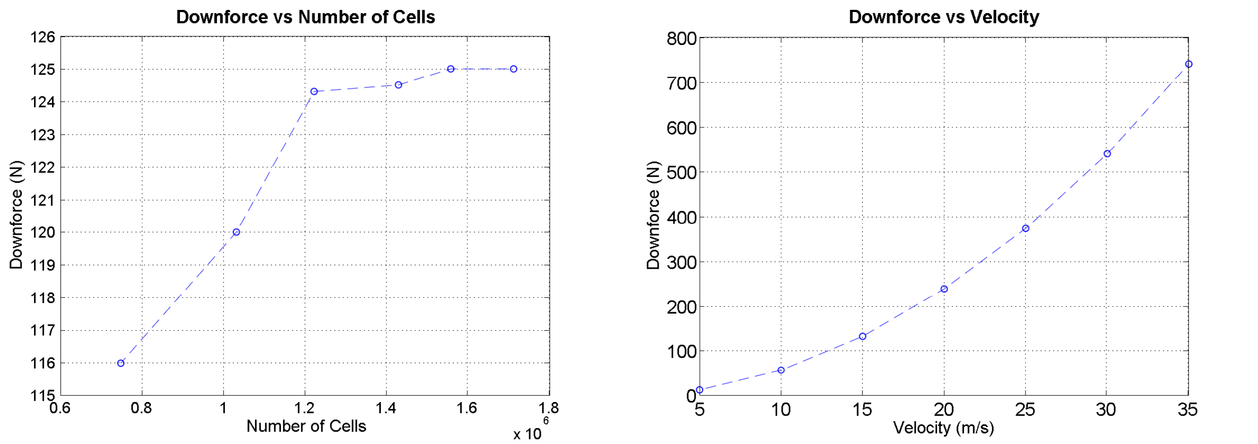
Figure 5: Example of mesh

Figure 4 describes the boundary conditions used to run simulations. In addition to the conditions described in the figure, rotating wheels were also used to better represent the local flow field. Prism layers were disabled on the moving floor to more accurately represent physical conditions as a boundary layer does not exist on the road during racing.

Mesh convergence

In order to best use our computational resources we conducted a mesh convergence study to investigate the variation in downforce values to differing mesh sizes. Figure 6a shows this relationship and it can be seen that a mesh size greater than 1.2 million cells produces a fairly static value of downforce. Therefore in order to reduce convergence times mesh sizes around 1.2 million cells were used throughout the project.

In addition to this, simulations were also completed using varying velocities. The results obtained, as depicted in Figure 6b perfectly represents the physical relationship between downforce and velocity. Drag vs velocity plots also resembles the same relationship hence verifying the theoretical hypothesis of a parabolic relationship.



(a) Variation of downforce with changing mesh size **(b)** Parabolic relationship between doenforce and velocity

Figure 6: Mesh convergence study

The results of this study were deemed to be satisfactory and therefore the mesh settings and size were used in our analysis for the rest of the project.

Optimisation strategy

Starccm+ was used to optimise all aspects of the front wing design. The design was optimised to produce the maximum amount of downforce for the lowest possible weight. Preliminary optimisation took the form of varying each parameter individually to analyse any improvements made. This allowed us to asses the independent effect of each parameter. A method like this also highlighted any sensitivities to our design, allowing the most influential parameters to be evaluated. In the section below, a detailed summary of results has been included. Upon identification of most influential parameters a two parameter study was carried out in order to find a better estimate of the optimum. Simulations performed in the initial study were used to populate the two parameter space. Table 3 depicts the various parameters that need to be optimised.

Table 3: Optimisation parameters

Parameters	
Ground clearance	Slot gap (x)
Main plane angle of attack	Slot gap (y)
First flap angle	Main plane chord
Second flap angle	Flap chord

Note that in this section the downforce and drag values being refereed to correspond to a half wing subjected to an inlet velocity of $15ms^{-1}$.

Single parameter studies

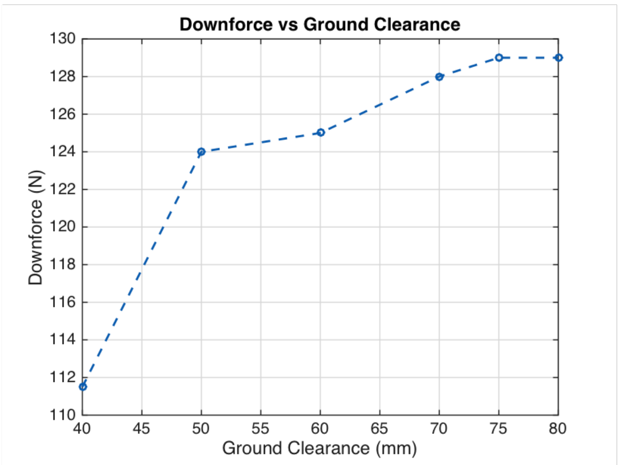
Due to the ease of conducting such a study and the time-scales involved in completing it, it was chosen to be our first protocol.

Ground clearance

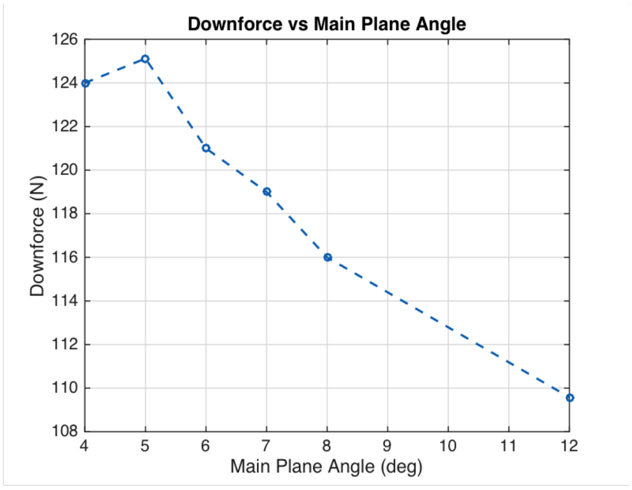
Gorund clearance was the first parameter that we analysed. It is defined as the shortest distance between the main plane and the ground. Looking at Figure 7a we can clearly see that downforce increases as clearance is increased for the values tested. At very low ground clearances, flow is accelerated to large velocities on the suction surface that result in low pressures, which benefits lift production. However the adverse pressure gradient when the flow is expanded is very large, causing separation near the trailing edge of the wing. This explains the loss in downforce at $40mm$. Although a large ground clearance would seem favourable, anything above $65mm$ would not be practical to implement. This is directly as a result of the height restriction set in front of the tyres. A large ground clearance would also mean that the flap in front of the tyre would be translated up. Therefore in-order to comply with the regulations the chord length of the flap would have to be reduced to impractical values. Hence, a ground clearance of $65mm$ was chosen. A ground clearance setting of $65mm$ provided a time advantage of 9 seconds in comparison to the unoptimised setting of $40mm$. This alone translated to a performance gain of 14 points in the endurance event.

Main plane angle

The main plane angle is defined as the angle of attack of the largest element of the front wing. Our design is untwisted across the entire span of the wing. Figure 7b shows the variation of downforce with changing angle of attack. Downforce reaches a maximum at $5°$, whilst dropping at both higher and lower angles of attack. This is slightly unintuitive as downforce is expected to increase at higher angles of attack. It is important to take into account that this drop in downforce is not related to stall. Rather, it is due to the fact that the main plane had to be truncated at high angles in order to comply with the regulations set in front of the tyre. Due to structural considerations the main plane needs to be continuous throughout the span. Therefore a larger α will mean that the chord of the whole main plane will have be reduced, not solely the region in front of the tyres. Reduced chord leads to reduced lift, hence explaining the drop in downforce at higher values of main plane angle. It is also to be noted that drag remained constant in these tests. Thus a main plane angle of attack of 5° was selected as the optimal value.



(a) Variation of downforce with ground clearance



(b) Variation of downforce with main plane angle of attack

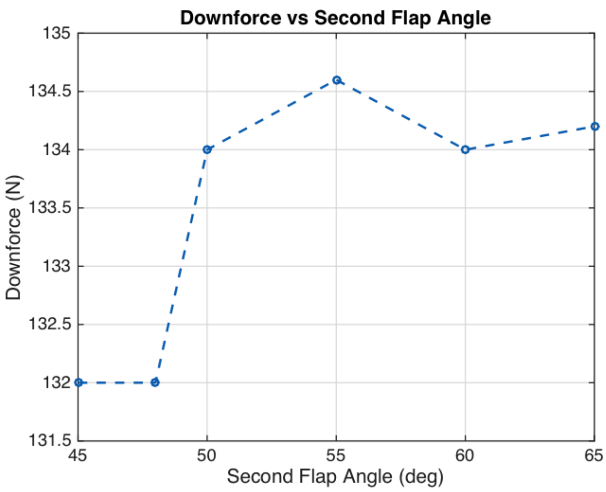
Figure 7: Single parameter study

First flap angle

As regulations didn't directly restrict this parameter, choosing the optimum was straightforward. Angles ranging from 23° to 45° were tested. The results showed the drag to remain nearly constant for the values tested whereas downforce was maximised at 39°. Hence a first flap setting angle of 39° was chosen.

Second flap angle

Figure 8 shows how downforce and drag vary for several flap setting angles. Although downforce increases with increased angle of attack, drag does too. Initially 55° was thought to be the optimum. However comparing the performance of flap settings of 55° and 45°, we found that a flap set at 45° reduced the total time by 1 second in the endurance event. This occurs as drag decreases by 7% when using the smaller angle. Although downforce does drop by 1.8% it proves to be insignificant.



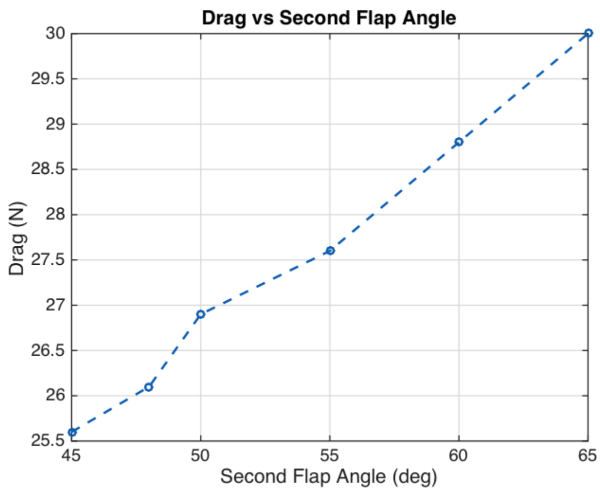
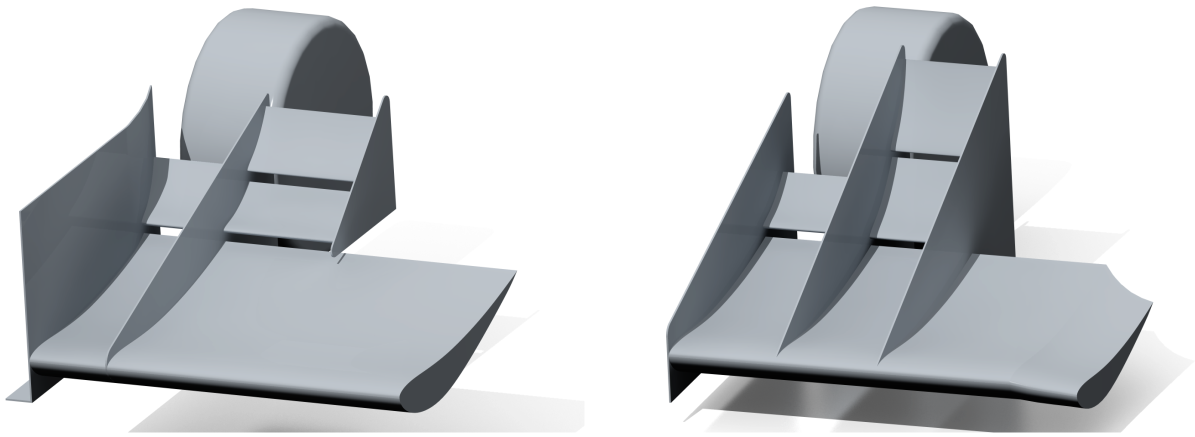


Figure 8: Variation of aerodynamic parameters with angle of attack



(a) Preliminary design

(b) Final design

Figure 9: Endplates and footplate

Tyre flap

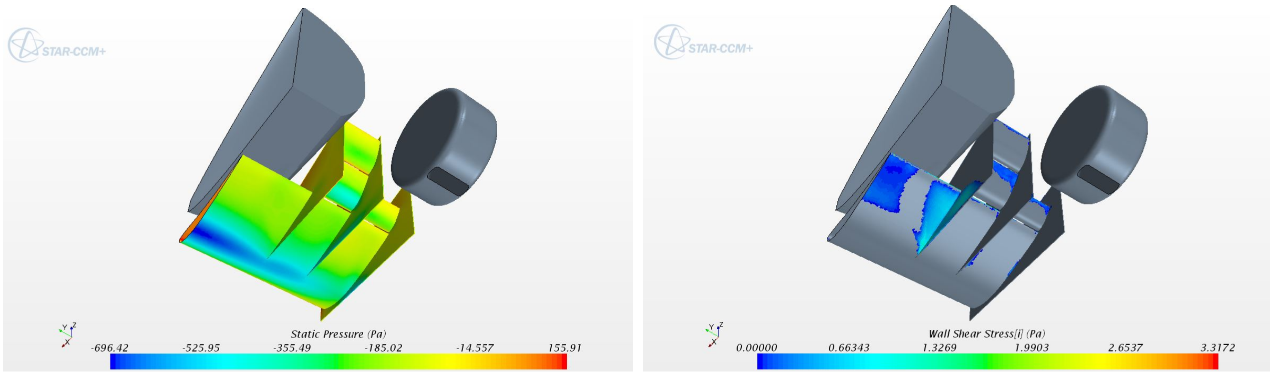
This is defined to be the flap that is directly in front of the tyres. As the rotating tyre behind the front wing acts as bluff body, it is essential to direct the flow away from the tyres to reduce drag. However, an overcorrection in the flow that would be caused by too high an angle of attack would also result in added drag. Simulations conducted showed that an angle of 40°minimised drag.

Endplate and footplate

Figure 9a and 9b show two iterations of the front wing, before and after the footplate and endplates were optimised.

The second and third elements in figure 9a are supported solely by the endplate they are attached to. At speeds when the lifting surfaces are producing substantial downforce, a large bending moment is applied at the joint between the elements and the flaps. These are essentially cantilevers and their structural capability was thought to be of concern. Therefore the endplate nearest the midspan of the wing was extended, allowing it to carry part of the load and therefore produce a structure with better load bearing capacity. However, this change alone caused the downforce to decrease by 5%. This was seen as a necessary compromise, which meant that the extended endplate is part of our final design. The decrease in downforce was as a result of separation on large portions of the main plane. The extrusion was not just limited to the pressure side but also extended below the main plane, as can be seen in figure 10a. This is a scalar representation of static pressure on the surface of the front wing. Note that the value being represented here is gauge pressure and is refernced to the ambient pressure. This figure views the front wing from the bottom up. It can be seen that a pressure gradient exists on the suction surface of the main plane. Peak suction pressure is found closest to mid span and gradually decreases along the spanwise direction. This is a direct result of the blockage caused by the nose and is evident by looking at pressure contour plots in appendix A.1a. As a result of the pressure gradient seen in figure 10a, crossflow is induced on the suction side of the main plane. The extruded endplate interferes with this crossflow resulting in separation on the suction side of the main plane. Figure 10b shows regions where wall shear stress is positive. (This indicates separated flow as the freestream is in the negative x direction) Large regions of positve wall shear stress next to the extruded endplate show separation, therefore explaining the loss of downforce.

Another noticeable change is the removal of camber from the endplate in the new design. Camber was initially introduced in the previous iteration with the aim of inducing outwash in the flow. This was done to direct the flow away from the front tyres, which act as bluff bodies and greatly add to drag. The introduction of outwash was successful in this objective and single handedly reduced drag by 7%. Although drag is reduced, the time improvement to a uncambered footplate is only 1 second



(a) Gauge pressure plot

(b) Wall shear stress in the longitudinal direction

Figure 10: Stand-alone CFD simulations. Note: Figure 10b only shows separated regions of the flow

for the whole endurance event. An improvement of this magnitude does not warrant the increased cost and manufacturing difficulty that a cambered endplate will add. Hence, the removal of camber is justified. As can be seen in figure 9b the footpate has also been removed in the final design. This follows similar reasoning as the predicted improvement translates to 1 point and yet again the margin of improvement is not substantial enough to justify the added difficulty in manufacturing.

Mass saving has also been done wherever possible, hence the endplates at the tip of the wing have been reduced in the final design. The lighter wing not only improves performance but also reduces material cost.

Two parameter optimisation

A two parameter study is a more robust way of finding the optimum as parameters are inevitably coupled. Populating a two parameter space will allow us to find a better combination of parameters aimed at improving performance. Given a much larger time span for the project, this would ideally be performed for all parameters. For the purpose of this project, a preliminary two parameter optimisation was done for x and y slot gaps.

Slot gaps

Literature review showed that a vertical slot gap and an overlap of 2% of the main element chord maximised the lift coefficient.[3] This translated to slot gaps of 9.5mm. Using this as an initial reference, we performed simulations near this value in-order to find the optimal slot gaps. Appendix A.4 outlines the result of these simulations. The largest performance was found to exist when the x and y slot gaps were set at 5mm and 10mm respectively.

Final design specification

The front wing spans a total of 1435mm while the leading edge of the wing is placed 700mm from the front of the tyres. The table below summarises the optimised parameter settings.

Table 4: Chord and angle of attack

Parameter	Chord (mm)	Angle of attack
Main plane	475	5°
Tyre flap	107	40°
First flap	156	39°
Second flap	156	45°

Table 5: Slot gaps

Direction	Slot gap (mm)
x	5
y	10

Ground clearance: 65mm

Structural considerations

During racing, the front wing is subject to various bending and torsional loads. At high speeds the downforce produced by the wing is considerable and it is therefore important to ensure that the wing as a load bearing structure is capable of withstanding such a load. Akin to the rest of the car, we will be using a high strength carbon fibre composite material to construct the front wing. Carbon fibre was chosen due to its high specific stiffness and strength. Table 6 provides the material properties used for all structural calculations performed.

Table 6: Material properties

Material property	Value
E_1	130 GPa
E_2	7.4 Gpa
ν_{12}	0.28
G_{12}	5.58 Gpa

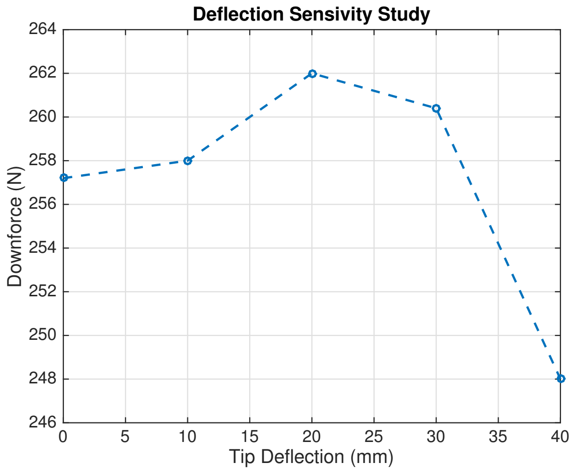


Figure 11: Variation of lift characteristics with tip deflection

Deflection sensitivity study

In order to consider the structural capabilities of our design, a sensitivity study was conducted to analyse the effect of bending deformation on the downforce produced by the front wing. Varying tip deflections in the range of 0 - 40mm were considered and the flow around each geometry was simulated in a CFD solver to produce figure . Using these results and the available ground clearance, a tip deflection of 30mm was deemed allowable. Anything larger would result in loss of performance and potential damage as the front wing collides with the floor.

A sandwich construction that included a pair of high strength carbon fibre laminate skins and low density foam core formed the front wing. The sandwich panel weighs little more than solely the carbon-fibre skins, however its flexural stiffness and strength is much greater. The larger the separation between the two skins and hence larger the thickness of the core material, the more favourable the discrepancy between the new material properties and the original. As weight is a limiting factor, sandwich structures have been proposed for the front wings.[4]

In order to calculate deflection at the tip, half the wing has been modelled as a cantilever. Conservative estimates have also been used wherever possible. For example, the weight of the wing along with the maximum downforce that can be produced has been placed at the tip of the wing to produce the largest possible moment. Although, not an accurate representation of the force distribution, an estimation such as this provides a safety factor in our calculations. In order to then verify if the structure is capable of withstanding the loads subjected to it we estimated a lay-up of $[0°/0°/0°/0°/\pm 45°/90°]_S$. Note that 0°fibres are in span wise direction of the wing, whereas the 90°fibres are parallel to the driving direction. 0°fibres have such notable presence as the largest loading condition is expected to bending about the longitudinal axis of the car. Fibres oriented at ±45°provide resistance to shear forces whereas fibres at 90°prevent bending about the axis parallel to the span. Once the structure of the laminate was decided we were then able to calculate the global stiffness matrix. Using laminate theory the deflection at the tip of the wing was calculated. Equations 1 and 2 were implemented in a Matlab script that calculated the D matrix and henceforth w, the tip deflection. The boundary conditions of zero curvature and zero deflection at the base of the cantilever were used to integrate equation 2.

$$M = D_{11}\frac{d^2w}{ds^2} \quad (1) \qquad w = \int\int\left(\frac{M}{D_{11}}\right)ds \quad (2)$$

Where M and s are the bending moment and the distance along the span of the wing, respectively.

Assuming the main plane is represented only by the wing box (leading edge and the trailing edge do not bear any load) a maximum tip deflection was found to be $7mm$. As this is well below the allowable tip deflection, the laminate was reduced by removing a total of four 0°plies. This helped reduce both weight and material cost. The new laminate still performed well and met the aforementioned deflection requirements. Table 7 summarises the results of the deflection study along with defining the final layup.

Table 7: Summary of results

Property	Value
Layup	$[0°/0°/\pm 45°/90°]_S$
Resultant deflection	$9mm$
Laminate thickness	$2mm$
Total weight	$7.6kg$
Material cost	£370

Although the deflection at maximum load is still well below the allowable limit, further reduction of the laminate is not recommended until a thorough analysis that includes torsional considerations is done.

Nose Attachment

Various methods to attach the front wing onto the nose have been investigated working along with the bodywork, structures and the global vehicle layout sub teams. (IRG08, IRG12 and IRG03 respectively) The initial concept involved attachment using four metal struts that were mounted directly onto the chassis. This allowed the load to be carried directly to the tubular frame. However when assessing the practicality of the solution, it was found that the length of the struts would have been approximately $700mm$, which is too large. The arrangement of struts would also have been problematic as the distance between the front wing and the chassis was large. The above reasons and the client's preference to adopt a different option meant that metallic struts were not utilised. Instead, the decision to use crushable pylons was pursued. Several iterations during the project involved a long nose that stretched close to the leading edge of the front wing. This allowed for a simple pylon structure that was normal to the floor. However, towards the end of the project the nose was shortened considerably following discussions with the bodywork and vehicle structures sub team due to structural concerns of having a long nose. This meant that slanted pylons, attached to the nose and the pressure side of the main plane had to be used.

The final design as seen in figure 3 has pylons in the shape of NACA-0012 aerofoils. The two pylons are glued directly using epoxy adhesive which can be conservatively approximated to have a failure stresses of $1MPa$ in both the axial and shear directions.[7] Rough calculations involving the contact area allowed us to conclude that the joint is stable even at the peak downforce condition.

This change to the nose in the final stages meant that the local flow field was affected, directly causing the downforce value to drop. The side view of the two nose geometries can be inferred from appendix A.1a. This figure along with the change in lift characteristics will be discussed in the next section.

Description of results

Due to the large computational expense of running full car simulations, optimisation of the wing was done with just the local geometry. This included the nose and the front tyre. Boundary conditions as described in the CFD section were also implemented. Table 8 shows lift characteristics as a result of simulations done on the final design.

Table 8: Percentage difference between stand alone and full car simulations

Property	Stand-alone (N)	Full car (N)	% Difference
Downforce	262	166	44.9
Drag	51	39	26.7

Full car simulation

Upon completion of individual optimisation of the complete aerodynamic package, the front wing sub-team set up and ran full car simulations.The results depicted in tables 8 and 9 were indeed very interesting as they showed downforce to decrease by a large amount. Downforce was expected to decrease in a full car simulation however a decrease of the order of $100N$ is not simply due to this. This is directly as a result of change in the nose geometry at the latter stages of the project. The new nose geometry significantly reduces velocities on the suction side of the main plane as the blockage effect is no longer as significant. The shorter nose is further back and this is visualised in figure A.1a. The figure below plots velocity below the suction side of the main plane for the two geometries of the nose against normalised distance in the longitudinal direction of the car. Refer to appendix A.2 for the location of these data points. It can clearly be seen that peak velocities are much greater in the presence of a longer nose, therefore resulting in lower pressures (Appendix A.3) and hence greater downforce. Note that the discrepancy of velocity at position zero is due to upstream influence of the nose.

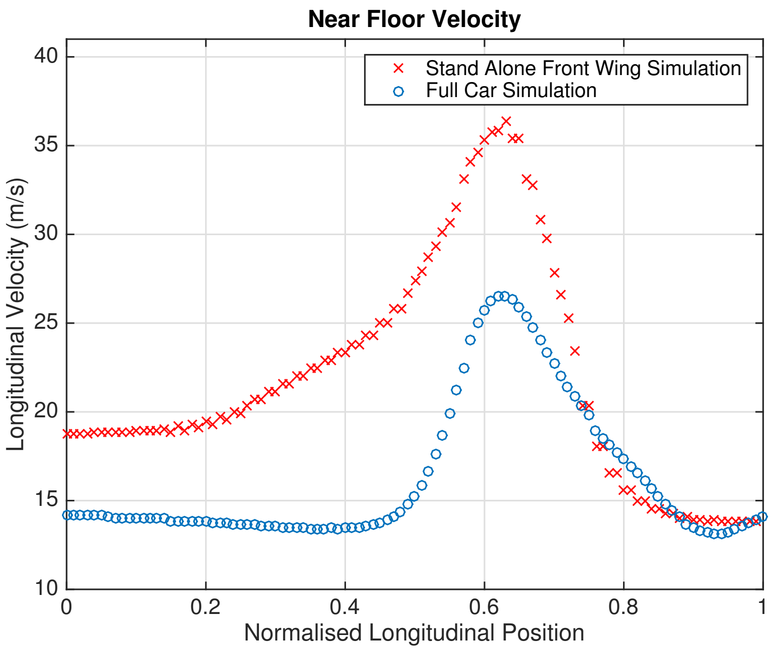


Figure 12: Near floor velocity. Refet to appendix A.2 for location of data points

Table 9: Description of results from the full car simulation

Full Car Simulation				
Contributor	Downforce (N)	Drag (N)	Weight (kg)	Material cost
Front wing	166	39	7.6	£370
Rear wing	170	54	9.1	£530
Diffuser	44	3	4.9	£170
Bodywork	-14	47	11.2	£625
Total	366	143	32.8	£1695

Performance estimation

The lap simulator was used alongside the vehicle performance subteam (IRG06) in order to asses the effectiveness of both the front wing and the aerodynamic package a a whole. Table 9 provides a breakdown of both the aerodynamic characteristics, weight and cost of the whole aerodynamic package. It was found that the front wing alone reduces the overall time of endurance event by 14 seconds. Such a performance gain for a cost of £370 proves that including the front wing is a cost efficient way to gain performance.

Cooling

Providing sufficient mass flow to the sidepods to allow for efficient cooling is also one of the responsibilities of the front wing. The cooling subteam (IRG11) specified a required mass flow rate of $0.68kgs^{-1}$ per sidepod. This was satisfied and the mass flow rate to each sidepod was equal to $0.76kgs^{-1}$.

Conclusion

The full car simualtion showed the front wing produced a total of $166N$ of downforce and $39N$ of drag. This was significantly lower than the $262N$ of downforce in stand-alone simulations due to the change in geometry of the nose in the latter stages of the project. Structural calculations allowed us to decide on the material layup to be used in a sandwich structure and let us conclude that the weight of the front wing would be equal to $7.6kg$. Final performance estimates using the lap simulator showed the front wing improved the time by 14 seconds in the whole endurance event. Considering the cost of the wing, this was deemed satisfactory.

Given a longer timespan for the project, the front wing sub team would have to work alongside the bodywork team in-order to re-optimise both the nose and the front wing to accommodate for the new design as the potential for better aerodynamic characteristics is present.

During the course of the project we also established that ground clearance and main plane angle of attack were the most influential parameters. Subject to available computational resources, an optimisation space including these two parameters would be populated in order to find an optimum setting that increases performance. The potential of using smarter optimisation strategies such as the optimal latin hypercube method should also be investigated. Although this might be computationally more expensive than the strategy used in this project, it could lead to a better performing front wing if implemented correctly.

The use of wind tunnels to correlate CFD data that was obtained would help improve estimations and provide more validation to design decisions made during the course of the project.

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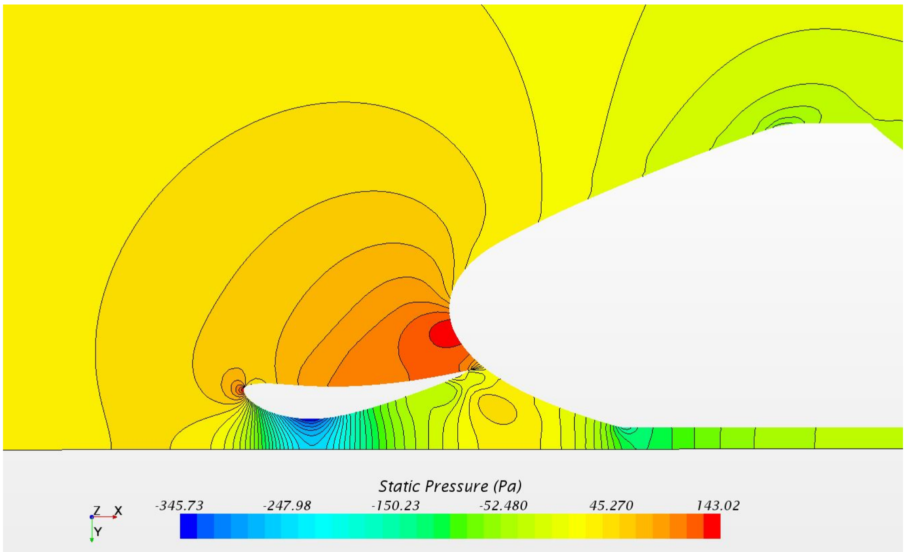
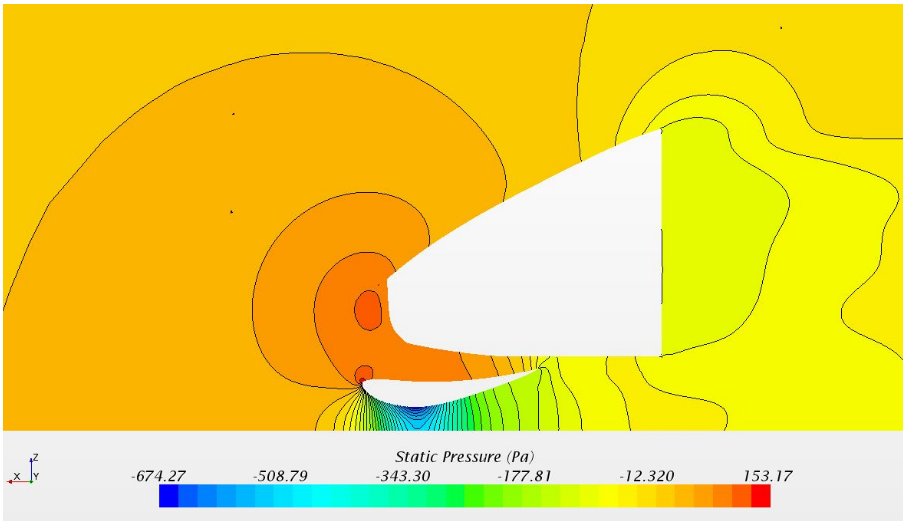
Appendix A

Comparison of nose geometries

Static pressure field using two nose geometries



(a) Pressure distribution on the latest nose geometry



(b) Pressure distribution on the long nose

Figure A.1: Scalar gauge pressure plots comparing two nose geometries (Side view)

Location of data points for figure 12

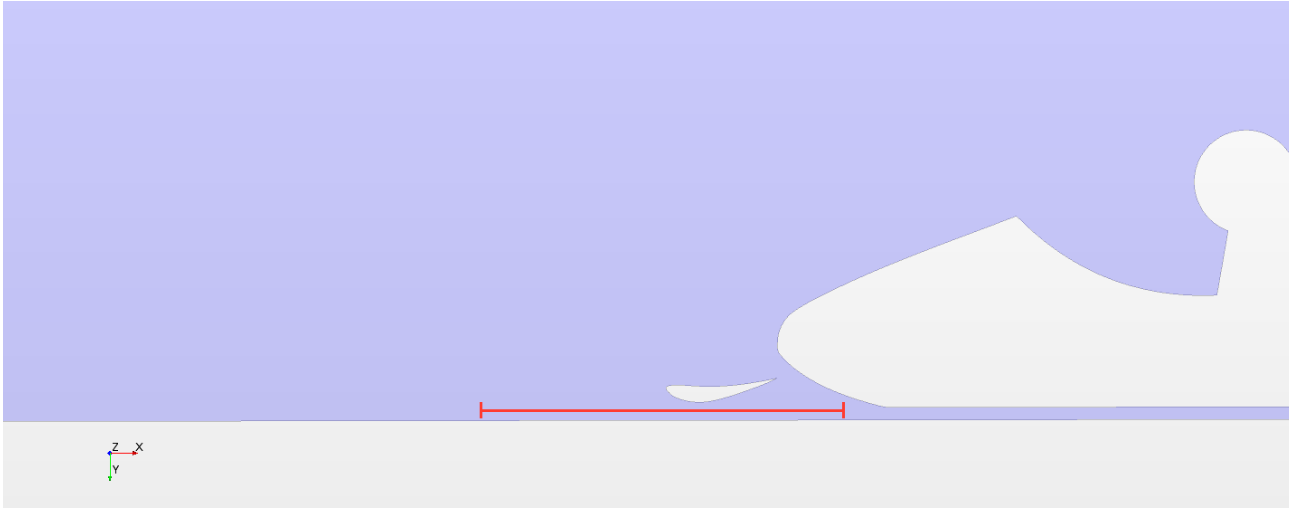
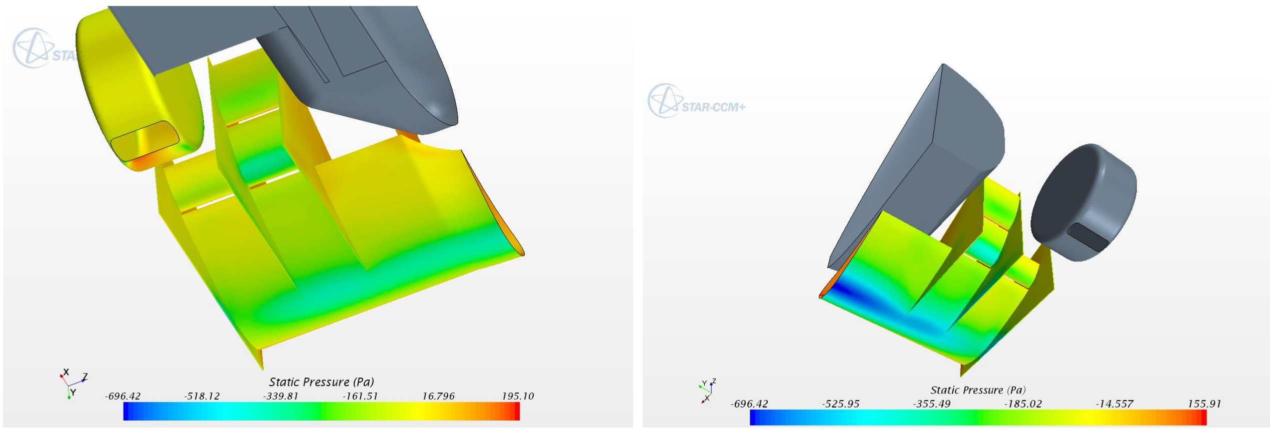


Figure A.2: Location of data points for figure 12

Comparison of static pressure on the suction side of the main plane due different nose geometries



(a) Gauge pressure plot - Short nose

(b) Gauge pressure plot - Long nose

Figure A.3: Scalar gauge pressure plots comparing two nose geometries

Note that the colour bars have been set to the same scale. Therefore pressure difference can be seen directly by looking at the colours.

Slot gaps

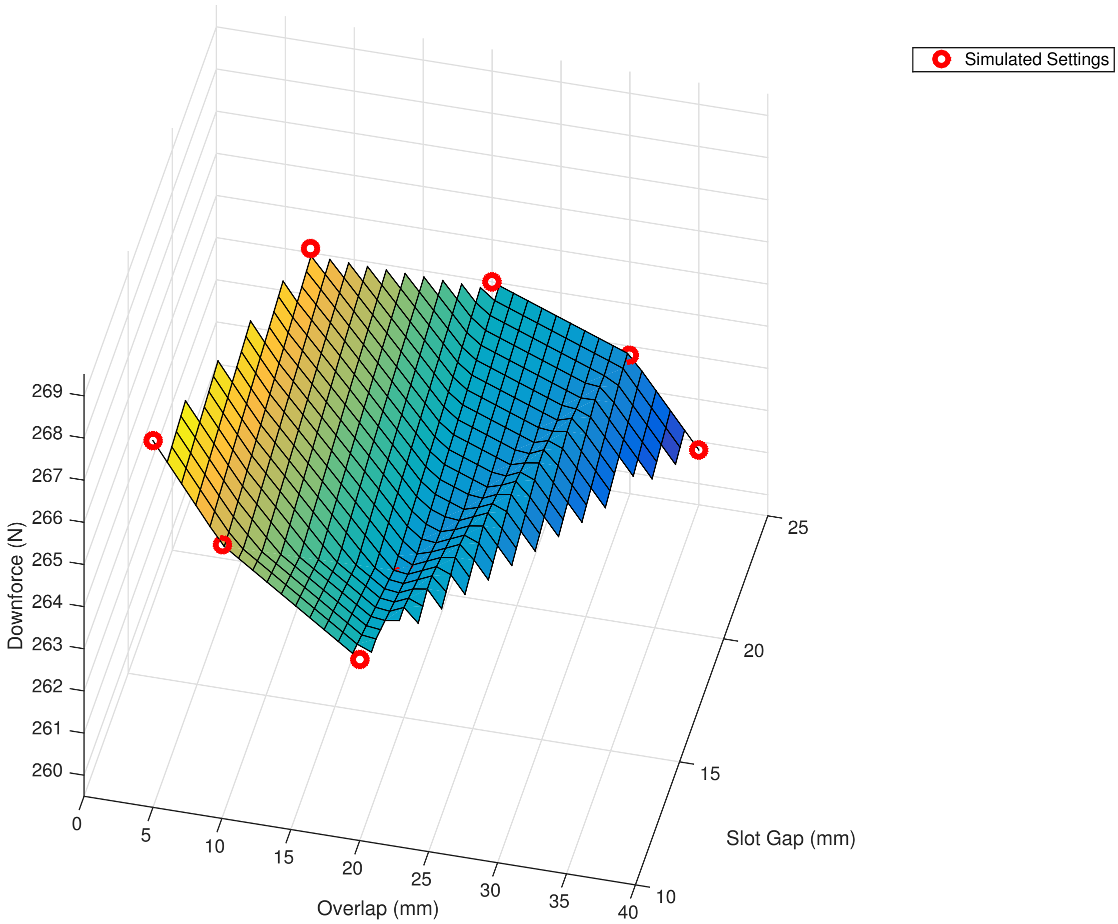


Figure A.4: Variation of aerodynamic parameters with changing slot gaps